Similarly, when the base unit is 9,

$9^1 = 9$ (odd power) $9^2 = 9 \times 9 = 81$ (even power)

$9^3 = 9 \times 9 \times 9 = 81 \times 9 = 729$ (odd power) $9^4 = 729 \times 9 = 6561$ (even power)

$9^5 = 6561 \times 9 = 59049$ (odd power) $9^6 = 59049 \times 9 = 531441$ (even power)

Thus, for base ending with 9, the unit digit after the expansion is 9 for odd power and is 1 for even power.

As we have seen in the earlier case, this rule is applicable, when the base is in the form of [(multiple of 10)+4] or [(multiple of 10)+9].

For example consider, 24^{12}

In this, unit digit of base 24 is 4 and power is 12 (even power).

Therefore, unit digit of 24^{12} is 6.

Similarly consider, 89^{21}

Here unit digit of base 89 is 9 and power is 21 (odd power).

Therefore, unit digit of 89^{21} is 9.

We conclude that, for base ending with 4, the unit digit of the expanded form is 4 for odd power and is 6 for even power. Similarly, for base ending with 9, the unit digit of the expanded form is 9 for odd power and is 1 for even power. Remember, 4 and 6 are complements of 10. Also, 9 and 1 are complements of 10.

Example 3.12 Find the unit digit of the large numbers: (i) 4^7 (ii) 64^{10}

Solution

(i) 4^7

Unit digit of base 4 is 4 and power is 7 (odd power).

Therefore, unit digit of 4^7 is 4.

(ii) 64^{10}

Unit digit of base 64 is 4 and power is 10 (even power).

Therefore, unit digit of 64^{10} is 6.

Try these

Find the unit digit of the following exponential numbers:

(i) 64^{11} (ii) 29^{18}

(iii) 79^{19} (iv) 104^{32}

Example 3.13 Find the unit digit of the large numbers: (i) 9^{12} (ii) 49^{17}

Solution

(i) 9^{12}

Unit digit of base 9 is 9 and power is 12 (even power).

Therefore, unit digit of 9^{12} is 1 .

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